Cristian Bautista

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Research Interests

Graduate Research Assistant at the **Center for Automotive Research (CAR), The Ohio State University**. My research focuses on **Autonomous Vehicles, multi-agent perception, and Visual Language Action Models (VLAs)**. I am particularly interested in developing **robust multi-modal perception systems** for safety-critical autonomous driving in complex real-world environments.

Education

The Ohio State University, Columbus, OH — May. 2026 – May. 2030
PhD. in Electrical and Computer Engineering — GPA: -.- / 4.0
Advisors: Prof. Qadeer Ahmed, Dr. Ekim Yurtsever

The Ohio State University, Columbus, OH — Aug. 2024 – May. 2026
M.S. in Electrical and Computer Engineering — GPA: 3.6 / 4.0
Coursework: Reinforcement Learning (A), Convex Optimization (A-), Autonomy in Vehicles (A), Robotics (A)
Advisors: Prof. Wei-Lun Chao, Prof. Qadeer Ahmed

Universidad Santo Tomás, — 2016 – 2021
B.S. in Electronic Engineering — GPA: 4.4 / 5.0
Highest Honors Graduation: Cum Laude & Laureate Thesis Award
Relevant Coursework: Computer Vision (A), Intro to Machine Learning (A)

Research Experience

Graduate Research Assistant — **The Ohio State University**
Research Projects — *Summer 2025–Present*

- **Collaborative 3D Perception:** Developed unsupervised methods for 3D object detection in multi-agent collaborative perception systems.
- **Full stack software for AVs:** Led full-stack autonomous vehicle integration, 2D/3D perception pipelines, SLAM, control interfaces, and CAN-based vehicle communication.
- **Learning-Based Decision-Making:** Designed hierarchical deep reinforcement learning agents for robotaxi missions in complex, real-world driving environments.

Buckeye AutoDrive Challenge — **The Ohio State University**
Team Captain & Mobility Innovation Leader — *Aug, 2024 – Jun, 2026*

- Led the end-to-end integration of a **Level-4 autonomous vehicle stack** with safety-critical and robust behavior for real-world deployment.
- Designed an optimized **multi-modal sensor suite** for autonomous driving under adverse weather conditions: (***Awarded***).
- Collaborative driving approach to mitigate misdirection from traffic agents; **Oral presentation** at *SAE WCX 2025* (Detroit, MI).

Universidad Santo Tomás — **Tunja, Colombia**
Plum Selection System (Undergraduate Thesis) — *2019–2021*

- Deep Learning approach to classify plums by morphology/visual features; optimized via data augmentation and Grad-CAM.
- Real world robotic system implementation to plum selection using Deep Learning.

Publications

When the City Teaches the Car: Label-Free 3D Perception from Infrastructure
Cristian Bautista*, Zhen Xu*, Jinsu Yoo*, Zanming Huang, Tai-Yu Pan, Zhenzhen Liu, Katie Z Luo, Mark Campbell, Bharath Hariharan, Wei-Lun Chao. *(*equal contribution)*
ECCV 2026 (*Under Review*) **Best Poster Award, X-Sense Workshop, CVPR 2026**
[Link: ArXiv Version]

SmartCityZero: Label-Free 3D Perception through RSU-to-Vehicle Collaboration
Cristian Bautista*, Zhen Xu*, Jinsu Yoo*, Zanming Huang, Wei-Lun Chao.
ICCV 2025 *Workshop X-Sense*

A Plum Selection System that Uses a Multi-class Convolutional Neural Network (CNN).
Cristian Bautista*, Yesid Fonseca*, Camilo Pardo, Carlos Parra.
Journal of Agriculture and Food Research (2023).

Plum Selection System using Computer Vision.
Cristian Bautista*, Yesid Fonseca*, Camilo Pardo.
IEEE ANDESCON (2020)

Teaching Experience

Universidad Santo Tomas — **Tunja, Colombia**
Teaching Assistant – Fourier Analysis — *Fall 2019, Spring 2020*
Give lectures, held office hours and assist the laboratories over 50 undergraduate students. Topics included Fourier series, Fourier transforms, and applications in **signal processing**.

Industry Experience

Cummins Inc. & Center for Automotive Research — **Columbus, IN & Columbus, OH**
Research Project — *Summer 2025*
- Contributed to regulatory compliance & functional safety alignment for automotive embedded systems.
- Supported AUTOSAR-based software architecture and requirements traceability workflows (ALM).
- Assisted in verification/validation of ISO 26262 safety-critical requirements.

Technip Energies — **Bogotá, Colombia**
Instrumentation & Automation Specialist I (Full-time) — *2022–2024*
- Led procurement and integration of pressure/temperature instrumentation for an European biofuels plant.
- Collaborated with global suppliers to evaluate, select, and validate industrial instrumentation.
- Designed process and pneumatic hook-up diagrams for Industrial sensors and air distribution systems

Awards & Recognition

2026: **Best Poster Award** — *X-Sense Workshop, CVPR 2026*
Awarded for the paper *When the City Teaches the Car: Label-Free 3D Perception from Infrastructure*.

2026: **AutoDrive Challenge II Year 5 Awards** — *Buckeye AutoDrive*
As **Team Captain**, led Buckeye AutoDrive to award-winning results across multiple categories: **3rd Place** in *Mobility Innovation* (main leader), **2nd Place** in *MathWorks Simulation Challenge*, **3rd Place** in *System Safety*, and **3rd Place** in *Overall Static Events*.

2025: **Promoted to Team Captain** — *Buckeye AutoDrive*
Leading a team of 50+ undergraduate and graduate students toward developing a fully autonomous Level 4 vehicle.

2025: **3rd Place, Mobility Innovation** — *AutoDrive Challenge II Year 4*
A prestigious Autonomous Vehicles competition, sponsored by SAE International and General Motors.

2023: **Colfuturo Scholarship** — *Ministry of Science Colombia*
Prestigious award supporting graduate studies abroad.

2022: **Opportunity Funds Program** — *U.S. Department of State*
Fellowship for high-potential students pursuing graduate studies in the U.S.

2021: **Cum Laude & Laureate Thesis Award** — *Universidad Santo Tomás*
Highest undergraduate honors.

2018–2019: **Merit-based Scholarship**
Awarded for outstanding academic performance.

2018: **International Exchange Scholarship** — *Universidad Santo Tomás*
Awarded for outstanding academic performance to represent the university abroad.

Technical Skills

Programming: Python, C++, MATLAB, ROS2, Git, Bash

Machine Learning & Deep Learning: PyTorch, TensorFlow, OpenCV, Open3D.

Perception & Sensor Fusion: LiDAR, Radar, Cameras, V2X; multi-sensor fusion, calibration, and synchronization.

High-Performance Computing: OSC: Slurm, multi-node GPU/CPU jobs, CUDA.

Languages: Spanish (Native), English (Full Proficiency).